

TCP/IP Model-IP Addressing - I

Module 09

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hello class today we are going to talk about cyber security as the name says security the first thing which comes in mind that how I'm going to secure my computer system now when we talk about security the first thing which we need to understand how the computer network works so we are going to start today with the introductory session on computer networks as all of we know that whenever a data has to transfer from one machine to another machine it is going to go from application which is going to talk to the user to the physical layer so let's first start with TCP IP model of networks so here as you can see there are seven layers in total which starts from application layer and you can memorize it by a phrase called

all people seem to need data processing now there is a client machine and there is a server machine server machine will also have the seven layers all people seem to need data processing now when we talk about a practical model the first three layers which are called application layer presentation layer and session layer these layer are combined together and this is called simply application layer so on that application layer we basically talk to a server using a browser so browser is an application which is basically interface between user and the Machine now here in the browser we are going to type address to which the site we are going to go to for example if we say www .

google.com so this is the address we have typed in the address bar now when these layers are going to talk to each other the conversation between the layer happens vertically down so we cannot have a horizontal communication so that means the data will travel from application layer to the transport layer and from the transport layer it will go to the network layer and from the network layer data link layer and finally it will go to the physical layer so that means the data link layer and physical layer are the two layers which are basically dependent upon the network type like whether I am using a Wi-Fi network or I am using a wired Network so when we transport the data from an application layer the data is called

payload that means I'm going to send a payload called www dot google.com so this is the payload which is going to go to the next layer now the next layer is called transport layer now on the transport layer what happens there are many things which happened at a transport layer which is governed by TCP and UDP protocols so there are two protocols prevalent on

this layer one is called TCP another is called UDP now the thing comes in mind basically what is a protocol so protocol is set of instructions or you can say it is set of rules which need to be followed in order to respect a particular line of action so that means we either choose a TCP or a UDP protocol most of the realtime applications whether we are going to

talk like www which is basically a HTTP related application or we have a mailing application all these applications are going to follow a TCP layer as a protocol so TCP is the default protocol for all realtime applications now inside TCP there are many things which will happen which we are going to explore in the later classes but as a introductory class we need to understand that there are two things basically which happens at TCP layer one is called The Source port address and another is called the destination port address now what this addresses basically are these addresses will identify a particular application running on the server so that mean these are the identifiers which will tell about applications running at those layers

like if HTTP application is running it is going to respect a number which is called Port 0 so Port 0 is recognized as an HTTP application now these are the standard application which will run on this particular machine now because we were accessing www.google.com so that means we are going to go to HTTP P protocol and this will listen at Port number 80 so that means from a payload when it comes to the transport layer this will be called as a segment so inside the segment we are going to add two things one is one will be the payload another will be the source port number and destination port number now the destination port number will be 80 by default because we are going to access it through a HTTP protocol so the

port number 80 is default for HTTP applications but what about the source port number from where the data is going to get originated now this Source port number is any random number which is above to 4 port numbers so basically a port number is 16bit entity that means if we say 2 to 16 this comes out to be 65,536 ports so these many application theoretically can run on one single machine now out of this we are going to bind a particular application to a port number like HTTP will get binded to port number 80 FTP will get binded to two port numbers 20 and 21 whereas SMTP gets binded to port number 25 similarly SSH will get binded to port number 22 tnet gets binded to port number 23 and these are the standard applications

which are being used across the world for accessing various kind of applications now when we talk about the source port numbers now the source port number cannot be any port number which is less than 1024 because these are standard port numbers so your computer is going to choose any number above 1024 let's say it chooses 4444 as the source port number now your segment will now look like 4444 which is your standard port number randomized for transport layer and then we have port number 80 which is default port number for HTTP application and in between there will be a payload which has come from the application layer so now the segment is going to go to the next layer which is called Network layer now Network layer is responsible

for giving addresses to the computers like client computer and server computer so the client computer will also have some address and server computer will also have some address now these addresses are given in a form which is called dotted decimal notation now this dotted decimal notation is called IP address now the size of the IP address is 32bit and we are talking here about IP version 4 so IP version 4 has 32bit address that means in totality I can have 2 raised to 32 unique addresses generated by ipv4 now out of these ipv4 2 raised to 32 addresses there are certain addresses which are being reserved which are called class D and E addresses rest of the addresses are used by our computers which are called Class A address Class B address

Class C address let's first of all talk about these addressing mechanisms now IP address basically is X.Y.Z.A now where each X or Y or Z or A will go from 0 to 255 so this will take eight bits now each OCTET is going to take 8 Bits so total we have 32 bits now the addresses will start from zero and it will go up to 255 for each octet so there are classes which are defined which is called Class A address Class B address Class C address and then finally D and E class now Class A how you will distinguish that particular IP address belongs to class A now you need to see that IP address is basically divided into two parts one part is called Network address and another part is called host address now the network address and host

address which part of IP address will be called as a network address and which part will be called as a host address it all depends upon something which is called subnet mask so subnet mask is very important in order to know which portion of IP address will be a network and which portion will address hosts now each of the classes like Class A address there is a default subnet mask now this default subnet mask is 255.0.0.0 so the addresses which have this subnet mask will be called as Class A address now that means the first OCTET if the value is from 0 to 126 these addresses will fall under category A address so let's take an example if I have addressed 10.10.10.10.

10 now by default if you see the first OCTET it lies between 0 and 126 that means this is a class A address now in the class address the subnet mask is 255.0.0.0 this is the default that means the first OCTET 8 Bits will represent Network and the three OCTETS that means 24 bits will represent host that means in a class A there will be 2 raised to 8 networks and each network will have 2 raised to 24 hosts so that means Class A networks are the largest networks which are bigger networks there will be millions of machines which can be in one of the class A networks then we move to class B now in class B the default subnet mask is 255.255.0.0 that means the 16 bits will represent Network and 16 bits will represent

host so the class B networks are 2 raised to 16 networks and each network will have 2 raised to 16 hosts so these are pretty big networks that means each network will have 65,536 hosts it's a huge number then we come to class C now Class C has a default subnet mask of 255.255.255.0 that means 24 bits will represent Network and 8 Bits will represent host so 2 raised to 8 number of host are available for each Network now 2 raised to 8 if we calculate is 256 hosts so

that means the smallest Network which we can create using Class C would house 256 hosts now let's take a practical example where I want to have 20 machines on a network now that means how many addresses are required if I want to address all the 20 machines we'll start

with the address one until address 20 so which addressing scheme will suit my network best so we have to choose Class 15 minutes, 1 second C because if we choose Class A we'll be wasting huge number of IP addresses so that mean there is a huge wastage which will happen if we use Class A because Class A has two ra to 24 hosts per Network so we will be going to choose Class C so Class C is going to give me 256 that means I am only wasting 236 IP addresses but if if you can see this is a huge wastage again it's a pretty huge wastage so what we need to do in such cases let's take an example

192.168.1.0 let's take this as an IP address and I want to address 20 machines with this IP address so the first address which I'm going to allot to my machine will be 192.168.1.1 the next machine will be 1.2 1.3 and it will go up to 1. 20 so these are the machines which will be allocated using this IP address so each of this machine will be having the same network so there are two things which are very important when the machines are going to talk to the another Network number one machine should be part of the same Network and second they should have something which is called a broadcast address so let's say there is a machine 192.168.1.2 it wants to talk to

192.168.1.2 so this mechanism will be called as unicast communication because a single machine is going to talk to another machine but least let's take an example

192.168.1.2 wants to talk to all the machines in its Network all the machines in the same network now first thing which we need to know what is the network of 192.168.1.2 let's find it in order to find a network we need to take an IP address and then take a logical and logical and operation of this with the subnet mask that means 192.168.1.2 is to be logically and with 255.255.255.0 so if we'll take a logical and that means I need to convert all these values into binary and then take the logical and so if we are going to take a logical and of anything with zero so this will be zero rest of the things will remain same so my network address comes out to be 192 .

16810 so this is the network address now what about the broadcast address broadcast address is when the all of your host bits are made one all host bits are made one that means this will be 192.168.1 dot 1 one1 1 one1 in binary so this comes out to be 255 so that means for this IP address 192.168.1.2 the network address is 192.168.1.0 and the broadcast address is 192.168.1 255 so that means for each network if the network has to talk to some other network we need basically two addresses one will be called as Network address and another will be called as broadcast address and if I wish to talk to another Network I'll require another special address which will be called as a gateway address so if I want to address 20

machines and these 20 machines are going to do a unicast communication broadcast communication and they also want to talk to some other network which is a different network then we'll require three addresses one will be Network address second will be broadcast address and third will be gateway address so these three addresses are required for each Network whenever a network wants to talk to the another Network so in our example 192.168.1.0 is the network address 192.168.1.255 is the broadcast address and you can choose any address to be chosen as address for your gateway so Gateway is nothing but it's a router which is going to talk to the another Network so if you want to follow a standard the standard say the last IP address of your network should be given

to the Gateway so the gateway address will be 192.168.1.254 that mean the formula is $2^n - 2$ to nus 2 hosts per Network so in our case the value of n was 8 so $2^8 - 2$ which comes out to be 254 hosts in class c network so each class c network will have 254 hosts and out of these 254 host one particular address will be given to the Gateway so that means my network is having 253 live hosts as per our example we needed only 20 addresses so that means finally we have 233 addresses getting wasted so this addressing scheme which is called classful addressing scheme was introduced in order to cut down wastage by classful addressing scheme so we have two schemes which are available now one is called classful which will use Class A address

Class B address or Class C address class D address is reserved for multicasting we'll talk about multicasting in the later lectures and Class E is not used this is a reserved class so we are left only with three classes Class A Class B and Class C but these classes gives us wastage so we are going to adopt another mechanism which is called cidr notation classless inter domain routing now this scheme is used in order to cut down the wastage so we are going to cut down the Wast stage let's see in our example how we can use cidr in order to cut down the wastage so we require 20 addresses so in order to address 20 machines how many bits I will require I will require five bits how five bits like we have a chart 1 2 4 8 16

32 now we'll give number of bits one bit 2 three four and five so we'll require five bits in order to address 31 machines $16 + 4 + 8 + 2 = 30 + 1$ so 31 machine can get address with five bits so that means I don't need to consume whole of the 8 bits of the host address I'll only consume five bits out of this and with the five bits I can address 31 machines so I require only 20 so that mean the wastage will be only 11 addresses so the cidr notation is going to help me in order to cut down this wastage now this kind of notation is called subnetting a network so we are going to subnet a network so that we can have less number of ips getting wasted so in our case 192.168.1.0 as a network address we are going to consume now only

five bits so that means three bits will be given to the network so if we are going going to write it in binary so it will be 1 1 1 1 2 3 4 5 so these five bits will be host bits and these three bits will be Network bits so what would be my Subnet Mask it will be 255.0.0.0 now these three I have to write it in decimal so this will be 1 2 4 8 26 minutes, 1 second 32 64 + 28

+ 255 so this is the value which is to be given in order to cut it to the subnet mask so my subnet mask will now will be different so it will not be 255.255.0.0 but it will be something else in the next stock date so cidr notation gives me a privilege so that less number of ips get wasted so we are going to take some examples in the next class where we are going to find out

subnets and even we'll talk about something which is called supernetting so super netting will be where the number of network will increase and subnetting is where the number of networks get decreased so that means we are going to play with the host bits and we are going to play with the network bits and this is going to cut down our vage of IP addresses so we are using IP V4 as our addressing mechanism so let's sum up whatever we have learned so far so that in the later class when we talk about subnetting and uh super netting we will be in the same line so we we started with an application layer so application layer has a payload this payload has to go vertically down two another layers precisely it will go to the transport layer and at the transport layer there

are two things which are going to get added one is called Source port number another is called destination port number and the protocol data unit which gets created at transport layer is called segment now segment will have destination port number which is a fixed entity and this destination port number is dependent upon the application from which we are going to receive the data like if we are going to receive a data from a browser application by default the destination port number will be 0 so we talked about these port numbers these standard port numbers are going to address standard applications and then with this segment the data will flow to the network layer now the at the network layer we

talked about two things one is the source IP address and destination IP address and in the IP addressing scheme we talked about about how to address the machines so machines can be addressed with the help of a classful IP addressing and classless IP addressing if we start using class full IP addressing lots of Ip V4 addresses get wasted as we saw in in an example we 192.168.1.2 Network and we wanted 20 machines to get address so we were wasting 236 IP addresses so we learned about something which is called classless interdomain routing so we are going to pick up some examples where it will be explained that how C is going to help us in order to cut down this so thank you very much for listening to this first introductory section on cyber security as a course so

in the next session we are going to explore about Network addresses and we are going to talk about Mac layer addresses which is called data link cayer addresses thank you